

Day 01 — Building the Logical Network Diagram

Mapping the full SOC home lab in draw.io before a single server is built — environment design, data flow, and the VPC blueprint.

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- DRAW.IO
- NETWORK DESIGN
- VPC
- ARCHITECTURE
- DATA FLOW

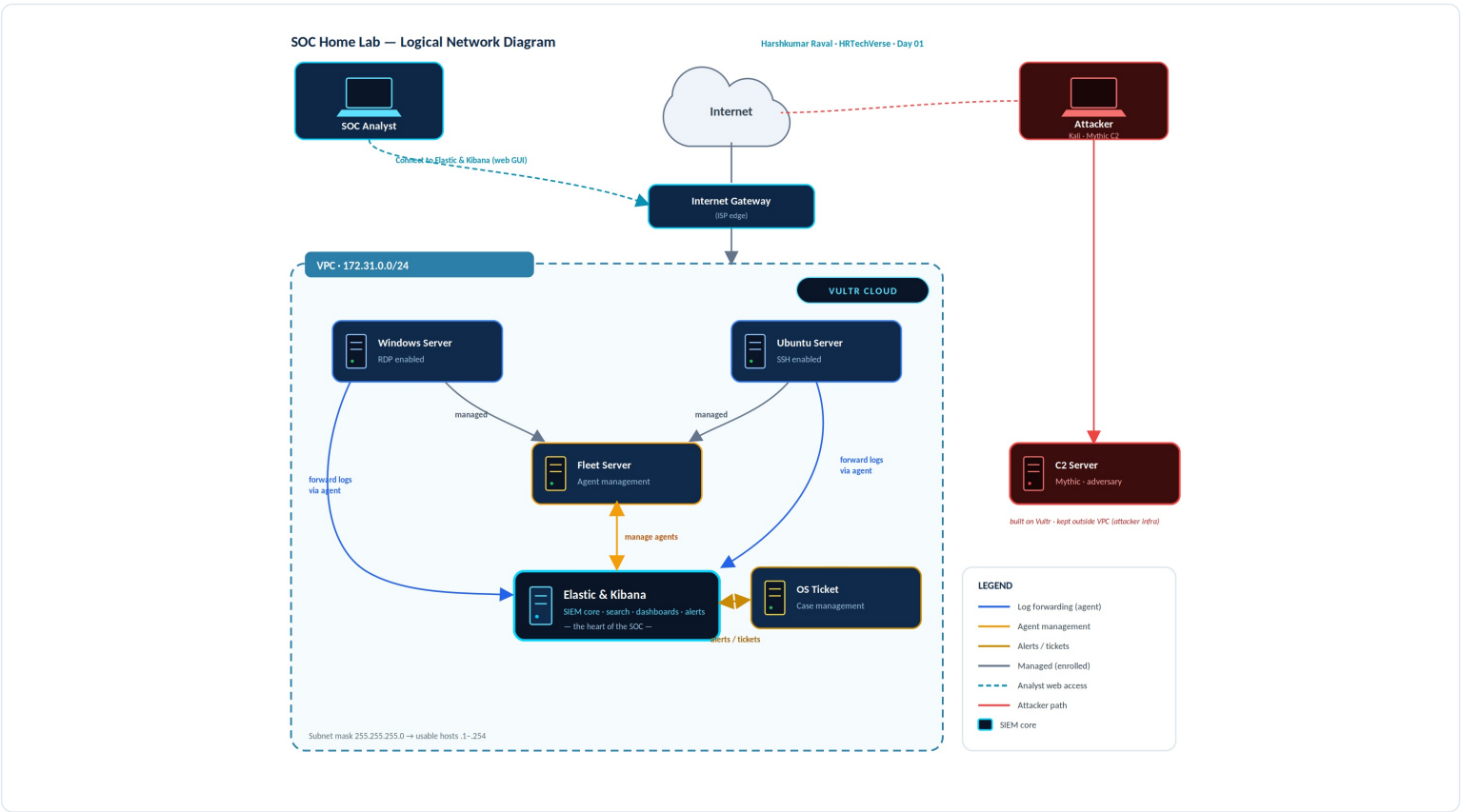
Objective

Before provisioning any infrastructure, a SOC analyst maps the environment. Day 1 builds a **logical network diagram** in [draw.io](#) that defines every server, how the network is segmented, and how data flows from endpoints into the SIEM. This diagram becomes the blueprint the rest of the 30 days is built against.

"In the beginning it might feel pretty useless, but trust me — this is a skill that will be extremely helpful in the future."

— STEVEN (MyDFIR)

The Lab Architecture



Logical network diagram — six Vultr servers, a segmented VPC, and the analyst + attacker access paths.

The Six Servers

All servers are provisioned on the **Vultr** cloud provider. Five sit inside the private VPC; the C2 server is intentionally placed outside as adversary infrastructure.

Server	Role	Purpose in the Lab
Elastic & Kibana	SIEM core	Stores, searches, and visualizes all logs; where alerts & dashboards live
Windows Server	Endpoint (RDP)	Monitored Windows host; target for RDP brute-force later
Ubuntu Server	Endpoint (SSH)	Monitored Linux host; target for SSH brute-force later
Fleet Server	Agent management	Enrolls and manages the agents that ship endpoint logs
OS Ticket	Case management	Ticketing system for alerts and investigations (Week 4)
C2 Server (red)	Adversary	Mythic command-and-control — placed outside the VPC

Network & Data Flow

Connection	Label	Meaning
Windows / Ubuntu → Fleet	managed	Endpoints are enrolled and managed by the Fleet server
Fleet ↔ Elastic & Kibana	manage agents	Fleet coordinates agent policy with the SIEM (bidirectional)
Windows / Ubuntu → Elastic	forward logs via agent	Endpoint telemetry is shipped into Elasticsearch
OS Ticket ↔ Elastic	alerts / tickets	Alerts become tickets; case status flows back (bidirectional)
SOC Analyst laptop → Elastic	web GUI	Analyst reaches Kibana over the internet via browser
Attacker laptop → C2	Kali · Mythic	Attacker operates the C2 from outside the VPC

VPC & Addressing

SUBNET PLAN

A **VPC (Virtual Private Cloud)** places all lab VMs on the same isolated private network. Range: **172.31.0.0/24** → usable hosts **172.31.0.1 – 172.31.0.254**, subnet mask **255.255.255.0**. An **Internet Gateway** (acting like the ISP edge) bridges the VPC to the public internet.

draw.io Skills Practiced

- Naming the diagram and toggling grid / shadow / sketch styling
- Searching the shape library (server, VPC, internet gateway, computer)
- Duplicating shapes with **Ctrl + D** and layering with **To Back / To Front**
- Drawing **bidirectional arrows** and recoloring lines to encode meaning
- Straightening connectors via the **Waypoint → Straight** option for a cleaner layout
- Saving the file: **File → Save**

Day 1 Checklist

- ✓ Open draw.io and name the diagram
- ✓ Place and label the six servers under the Vultr provider
- ✓ Draw the VPC, internet gateway, and internet
- ✓ Add the SOC analyst and attacker laptops
- ✓ Connect all servers and color-code the data-flow arrows
- ✓ Document the private network range and subnet mask
- ✓ Save the diagram and commit it to the repo

MY TAKEAWAY

The diagram isn't decoration — it's the mental model. Once the flow of "endpoint → agent → Fleet → Elastic → alert → ticket" is drawn out, every later day has a place to slot into. It also doubles as the single best artifact for explaining the lab to an interviewer in 30 seconds. Reps on logical diagrams compound.